

Exercise 2.1: Consider the feasible region defined by the constraints,

$$1 - x_1^2 - x_2^2 \geq 0, \sqrt{2} - x_1 - x_2 \geq 0, \text{ and } x_2 \geq 0.$$

For each of the following points, determine whether the point is feasible or infeasible and (if it is feasible) whether it is interior to or on the boundary of each of the constraints:

1. $x_a = (\frac{1}{2}, \frac{1}{2})^T$.

Solution:

Checking constraints,

$$1 - \left(\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 = 1 - \frac{2}{4} = \frac{1}{2} > 0$$

$$\sqrt{2} - \left(\frac{1}{2}\right) - \left(\frac{1}{2}\right) = \sqrt{2} - 1 > 0$$

and clearly $\frac{1}{2} > 0$. Thus x_a is a feasible point, and since we have strict inequalities on all the constraints we know that x_a is interior to all constraints (Defined in Chapter 2 p.44).

2. $x_b = (1, 0)^T$

Solution:

Checking all constraints,

$$1 - 1^2 - 0^2 = 1 - 1 = 0$$

$$\sqrt{2} - (1) - (0) = \sqrt{2} - 1 > 0$$

and since $x_2 = 0$ we know that x_b is a feasible point. Note that x_b is on the boundary of $1 - x_1^2 - x_2^2 \geq 0$ and $x_2 \geq 0$, and interior to $\sqrt{2} - x_1 - x_2 \geq 0$.

3. $x_c = (-1, 0)^T$

Solution:

Checking all constraints,

$$1 - (-1)^2 - 0^2 = 1 - 1 = 0$$

$$\sqrt{2} - (-1) - (0) = \sqrt{2} + 1 > 0$$

and since $x_2 = 0$ we know that x_c is a feasible point. Note that x_c is on the boundary of $1 - x_1^2 - x_2^2 \geq 0$ and $x_2 \geq 0$, and interior to $\sqrt{2} - x_1 - x_2 \geq 0$.

4. $x_d = (-\frac{1}{2}, 0)^T$

Solution:

Checking all constraints,

$$1 - \left(-\frac{1}{2}\right)^2 - 0^2 = 1 - \frac{1}{4} = \frac{3}{4} > 0$$

$$\sqrt{2} - (-\frac{1}{2}) - (0) = \sqrt{2} + \frac{1}{4} > 0$$

and since $x_2 = 0$ we know that x_d is a feasible point. Note that x_d is on the boundary of $x_2 \geq 0$, and interior to $1 - x_1^2 - x_2^2 \geq 0$ and $\sqrt{2} - x_1 - x_2 \geq 0$.

5. $x_e = (1/\sqrt{2}, 1/\sqrt{2})^T$?

Solution:

Checking all constraints,

$$1 - (\frac{1}{\sqrt{2}})^2 - (\frac{1}{\sqrt{2}})^2 = 1 - \frac{1}{2} - \frac{1}{2} = 0$$

$$\sqrt{2} - (\frac{1}{\sqrt{2}}) - (\frac{1}{\sqrt{2}}) = 0$$

and since $1/\sqrt{2} > 0$ we know that x_e is a feasible point. Note that x_d is on the boundary of $1 - x_1^2 - x_2^2 \geq 0$ and $\sqrt{2} - x_1 - x_2 \geq 0$, and is interior to $x_2 \geq 0$.

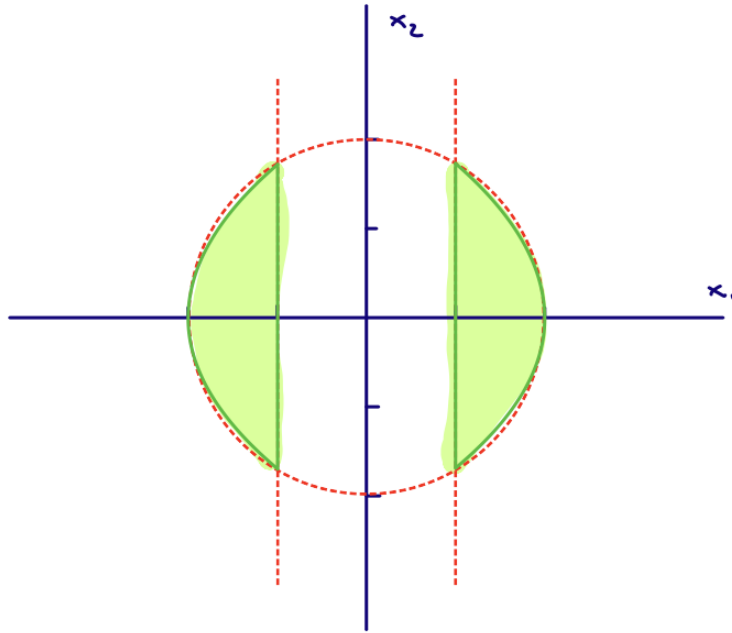
Exercise 2.3: Consider the problem,

$$\begin{aligned} &\text{minimize } f(x) = x_1, \\ &\text{subject to } x_1^2 + x_2^2 \leq 4 \\ &\quad x_1^2 \geq 1. \end{aligned}$$

Graph the feasible set. Use the graph to find all local minimizers for the problem, and determine which of those are also global minimizers.

Solution:

Consider the following plot of the feasible set.

Figure 1: Feasible set S denoted in green.

Since the objective function is $f(x) = x_1$ it's clear that there are a set of local minimizers when $x_1 = 1$, and $x_2 \leq \pm \sqrt{3}$. There is a single global minimizer where $x_1 = -2$, and $x_2 = 0$.

Exercise 2.4: Consider the problem,

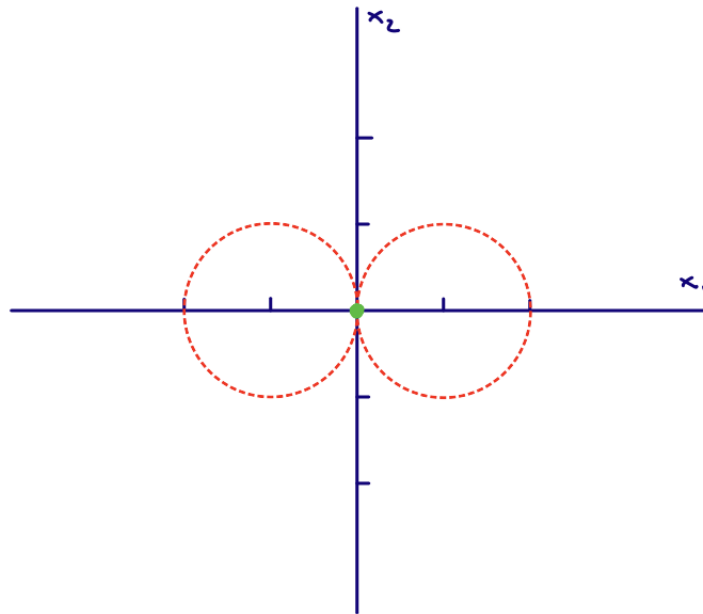
$$\begin{aligned} &\text{minimize } f(x) = x_1, \\ &\text{subject to } (x_1 - 1)^2 + x_2^2 = 1 \\ &\quad (x_1 + 1)^2 + x_2^2 = 1. \end{aligned}$$

Graph the feasible set. Use the graph to find all local minimizers for the problem, and determine which of those are also global minimizers.

Solution:

Consider the following plot of the feasible set.

Figure 2: Feasible set S denoted in green.

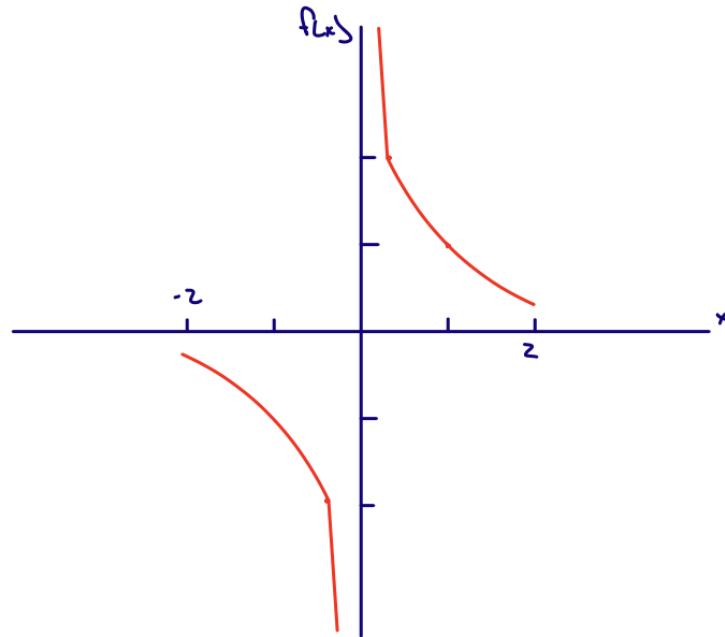


Since the feasible set contains only one point, namely $x_1 = 0$, $x_2 = 0$ it must be the global minimizer.

Exercise 2.5: Give an example of a function that has no global minimizer and no global maximizer.

Solution:

Consider the function $f(x) = 1/x$ on the interval $[-2, 2]$.

Figure 3: Objective function on feasible set $x \in [-2, 2]$.

Note that there is a local minimizer at $x = 2$ but clearly any global minimizer will be on the interval $[-2, 0)$. Since $f(x)$ is strictly decreasing and divergent as x approaches 0, it is impossible to construct an ϵ -neighborhood around any x_* such that $f(x_*) \leq f(x)$ for all $x \in [-2, 0)$.

Similarly any global maximizer must be on the interval $(0, 2]$. Since $f(x)$ is strictly decreasing and divergent as x approaches 0, it is also impossible to construct an ϵ -neighborhood around any x_* such that $f(x_*) \geq f(x)$ for all $x \in (0, 2]$.

Exercise Problem P1: Solve calcone by implementing a strategy for solving this and similar one-variable optimization problems on bounded, closed intervals. Describe the strategy in brief and well-written english. Your strategy will necessarily be iterative, and will not get the exact answer but only an approximate one. Discuss any issues about the general performance/success of your strategy, emphasizing how it might fail on problems of this type. Argue that your answer to this particular problem is given to at least 6-digit accuracy, and along the way visualize the function $f(x)$.

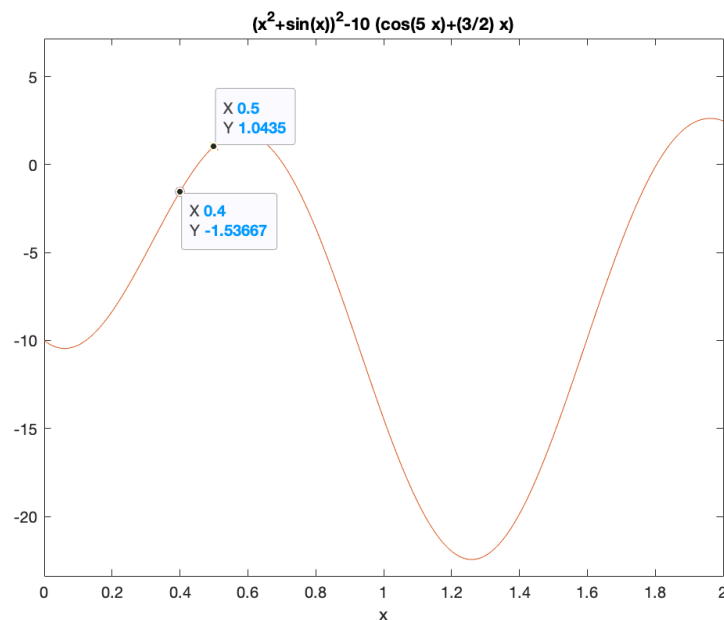
$$\text{minimize } f(x) = (x^2 + \sin(x))^2 - 10 \left(\cos(5x) + \frac{3}{2}x \right),$$

on the interval $S = [0, 2]$

Solution:

The strategy I implemented for solving this problem is the minimization analog to bisection root finding. Consider a function f on a closed interval $[a, b]$. Consider points $u, v \in [a, b]$ such that $u < v$. Compute $f(u)$ and $f(v)$. If $f(u) > f(v)$ then there must exist a local minimum on the set $[u, b]$. Similarly if $f(v) > f(u)$ there must exist a local minimum on the set $[a, v]$. We repeat this iteratively until the interval becomes essentially an ϵ neighborhood of a local minimum. Clearly a function could exploit this method, consider the following plot if we were to let $u = .4$ and $v = .5$ the method would converge to the local minimum near 0.

Figure 4: Bisection Counter Example



To counter this I propose we take several equally spaced intervals on $[a, b]$ small enough such that the function on any sub-interval $[x_i, x_{i+1}]$ contains very little change in concavity. The final code (looks like psuedocode;)

Code:

```
function [fmin, xmin] = bisection(f, a, b)
%% Recursive bisection function, takes an interval [a, b]
%% returns a local minimizer (fmin, xmin)
```

```

    if abs(f(b) - f(a)) < 1e-6
        fmin = f(a + (b - a)/2);
        xmin = a + (b - a)/2;
        return
    end

    %% Identifying bounds for sub intervals.
    u = a + (b - a)/3;
    v = a + (2*(b - a))/3;

    %% Recurse
    if f(u) < f(v)
        [fmin, xmin] = bisection(f, a, v);
        return
    end

    if f(v) < f(u)
        [fmin, xmin] = bisection(f, u, b);
        return
    end
end

function [fmin,xmin] = BisectionMethod(f, a, b, n)
%% Solution to calcone, bisection method applied to n sub intervals.
%% Initializing minimum vectors.
fmin = [];
xmin = [];

%% Applying bisection to n sub intervals
interval = (b - a)/n;
for i = 1:n
    [fminInterval,xminInterval] = bisection(f, a + ((i - 1)*interval), a + (i*interval)
    fmin = [fminInterval fmin];
    xmin = [xminInterval xmin];
end

%% Pulling smallest f(x), x pair of all sub intervals.
[y, index] = min(fmin);
x = xmin(index);
end

```

Console:

```

>> f = @(x) (x.^2 + sin(x))^2 - 10*(cos(5*x) + (3/2)*x)
f =
    function_handle with value:
        @(x)(x.^2+sin(x))^2-10*(cos(5*x)+(3/2)*x)

>> [fmin, xmin] = BisectionMethod(f, 0, 2, 100)

fmin =

```

-22.448628772724792

xmin =

1.259282253345150

We exit the recursion when the bounds of the current interval, say $[a, b]$ are smaller than 10^{-6} . Since we know the interval $[a, b]$ must contain a local minimum any point within $[a, b]$ will also be within 6 digits accuracy to the true minimum in $[a, b]$.

Exercise Problem P2: Solve fit. Follow essentially the same rules as the previous problem. Describe a strategy for solving this and similar problems. Discuss any issues about the general performance/ success of your strategy. Implement your strategy as Matlab code. Demonstrate 6 digits of accuracy. Plot the solution curve on the same graph as the data.

Solution:

Note that we are aiming to fit $\hat{c}$ of the function

$$g(x) = c_1 + c_2x + c_3e^{-5x},$$

Such that the sum of squares to our data points are minimized. Therefore for every y_i we seek to minimize e_i and generally the sum squared error $\frac{1}{2} \sum \epsilon_i^2$,

$$y_i = c_1 + c_2x_i + c_3 + \epsilon_i.$$

Consider the following system of equations,

$$\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_{11} \end{bmatrix} = \begin{bmatrix} 1 & x_1 & e^{-5x_1} \\ 1 & x_2 & e^{-5x_2} \\ \vdots & \vdots & \vdots \\ 1 & x_{11} & e^{-5x_{11}} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_{11} \end{bmatrix}$$

$$\hat{y} = X\hat{c} + \hat{\epsilon}.$$

Solving for $\hat{\epsilon}$ we get,

$$\hat{\epsilon} = \hat{y} - X\hat{c}.$$

Applying sum of square over two we get, what is essentially the given objective function.

$$f(\hat{c}) = \frac{1}{2} \hat{\epsilon}' \hat{\epsilon} = \frac{1}{2} (\hat{y}' - \hat{c}' X') (\hat{y} - X\hat{c}).$$

Expanding the right hand side.

$$f(\hat{c}) = \frac{1}{2}(\hat{y}'\hat{y} - \hat{y}'X\hat{c} - \hat{c}'X'\hat{y} + \hat{c}'X'X\hat{c}) = \frac{1}{2}(\hat{y}'\hat{y} - 2\hat{c}'X'\hat{y} + \hat{c}'X'X\hat{c})$$

Taking the gradient of our objective function in terms of $\hat{c}$,

$$f'(\hat{c}) = \frac{1}{2}(-2X'y + 2X'X\hat{c}) = -X'y + X'X\hat{c}$$

Setting our gradient, $f'(\hat{c}) = \hat{0}$ and solving for $\hat{c}$ we get the following system of equations,

$$\begin{aligned} -X'y + X'X\hat{c} &= 0, \\ X'X\hat{c} &= X'y, \\ \hat{c} &= (X'X)^{-1}X'y. \end{aligned}$$

Our procedure for solving this problem will continue by using in-place householder decomposition on $X'X$, forming $Q^*X'y$, and then solving for $\hat{c}$ via back substitution.

Code:

```
function b = inhousesolve(A, b)
% This function takes a square matrix A and a column vector b
% and solves A\b via inplace householder qr.

%Error Checking
[m, n] = size(A);
[brow, bcol] = size(b);
if m ~= n
    error('This function only accepts square Matrices')
end

if (bcol ~= 1) || (brow ~= m)
    error('This function only a mx1 column vector')
end

%Inplace Computation of R and W
A = [A; zeros(1,m)];
for k = 1:n
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%% This Block is the same as house.m %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    v = A(k:m,k); % v is (m-k+1) x 1 column vector
    v(1) = sign(v(1)) * norm(v,2) + v(1);
    v = v / norm(v,2);
    A(k:m,k:n) = A(k:m,k:n) - 2 * v * (v' * A(k:m,k:n));
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    A(k+1:m+1,k) = v;% store v in A
end
```

```
%Linear Solve by only referencing A
%Forming Q*b in place of b
    for k = 1:m
        b(k:m,:) = b(k:m,:) - 2*A(k+1:m+1,k)*(A(k+1:m+1,k)')* b(k:m,:));
    end

%Solving for x via backsubstitution in place of b.
b(m,1) = b(m,1) / A(m,m);
    for i = m-1:-1:1
        b(i,1) = (b(i,1) - A(i,i+1:m) * b(i+1:m,1) ) / A(i,i);
    end

end
```

Console:

```
%% Data Input/Setting Up System
>> x = linspace(0,1,11);
>> y = [4.914 3.666 2.289 1.655 1.029 0.739 0.393 0.090 -0.197 -0.721 -0.971];
>> f = @(x) exp(-5*x)
>> X = [ones(11, 1), x', f(x)']
```

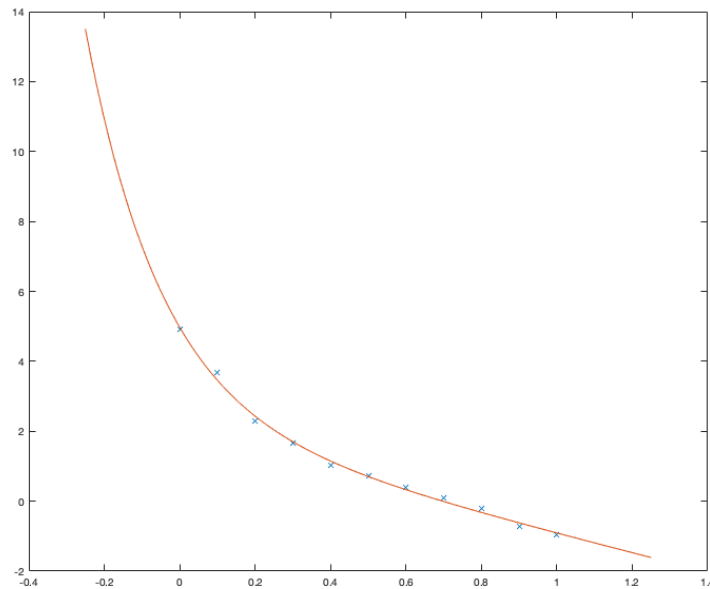
X =

1.0000000000000000	0	1.0000000000000000
1.0000000000000000	0.1000000000000000	0.606530659712633
1.0000000000000000	0.2000000000000000	0.367879441171442
1.0000000000000000	0.3000000000000000	0.223130160148430
1.0000000000000000	0.4000000000000000	0.135335283236613
1.0000000000000000	0.5000000000000000	0.082084998623899
1.0000000000000000	0.6000000000000000	0.049787068367864
1.0000000000000000	0.7000000000000000	0.030197383422319
1.0000000000000000	0.8000000000000000	0.018315638888734
1.0000000000000000	0.9000000000000000	0.011108996538242
1.0000000000000000	1.0000000000000000	0.006737946999085

```
%% Solving System
>> c = inhousesolve(X'*X, X'*y')
```

```
1.823144071637949
-2.752330574924139
3.148516264633391
```

```
%% Generating Plot
>> f = @(x) c(1) + c(2)*x + c(3)*exp(-5*x)
>> plot(x, y, 'X')
>> hold on
>> plot(linspace(-.25, 1.25, 100), f(linspace(-.25, 1.25, 100)))
```

Figure 5: Plot of $f(x) = 1.82 - 2.75x + 3.15e^{-5x}$ 

The flops for solving a triangular system is $O(n^2)$ and Householder decomposition is $O(n^3)$, so this method finds the optimal $\hat{c}$ in $O(n^3)$. (Reference NLA Trefethen, Bau Algorithm 11.2 p.83)

Exercise Problem P3: Solve Salmon. In fact this problem is embarrassingly simple to solve. Start by writing a few clear sentences stating and justifying the solution. Then visualize, in 3D and probably with pencil and paper, the set of feasible solutions; mark and label the solution on this plot. Also use a straightforward substitution to eliminate the equality constraint, and then re-visualize the feasible set and solution in 2D. Is this problem discrete? Is it fair to interpret it as continuous? Comment.

Salmon is a constrained minimization problem where the goal is to minimize,

$$f(x) = 2x_1 + 3x_2 + 4x_3$$

subject to.

$$x_1 + x_2 + x_3 = 22$$

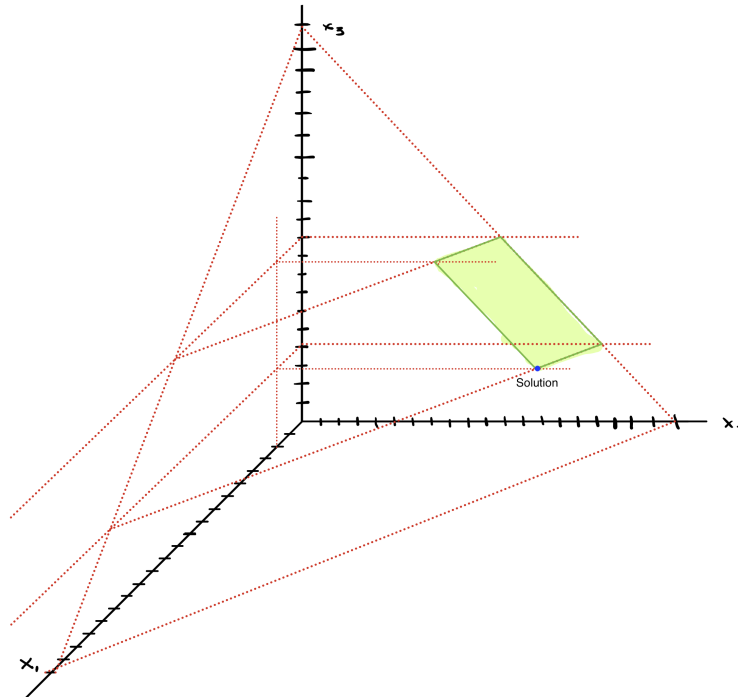
$$0 \leq x_1 \leq 2$$

$$0 \leq x_2$$

$$4 \leq x_3 \leq 10.$$

Solution:

Note that the total number of fish is 22 as denoted by the first constraint. Since x_1 contributes the least to increasing $f(x)$ it follows that we should maximize x_1 under the $0 \leq x_1 \leq 2$ constraint. Therefore $x_1 = 2$, and thus we have that $x_2 + x_3 = 20$. Since x_3 contributes the most to increasing $f(x)$ it also follows that we should minimize x_3 under the $4 \leq x_3 \leq 10$ constraint. Thus $x_3 = 4$ and we $x_2 = 16$. Shifting the values to lower x_1 or raise x_3 results in a larger $f(x)$ and thus $x = [2, 16, 4]'$ minimizes $f(x)$. Consider the following plot of the feasible set,

Figure 6: Feasible set S denoted in green.

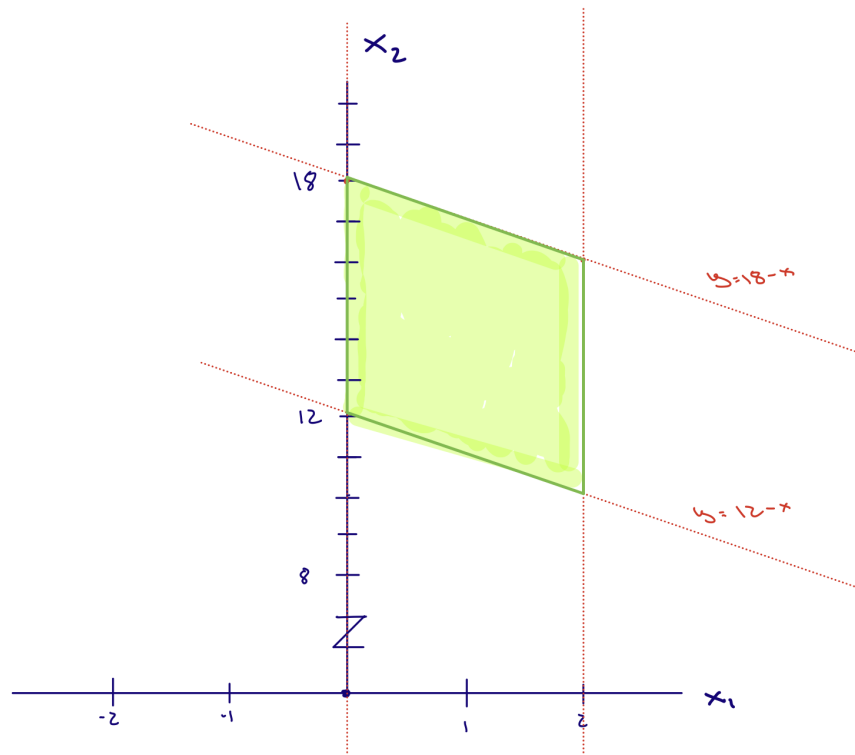
Solving the equality constraint for x_3 we get the following problem,

$$f(x) = 2x_1 + 3x_2 + 4(22 - x_1 - x_2) = -2x_1 - x_2 + 88$$

subject to.

$$\begin{aligned} 0 &\leq x_1 \leq 2, \\ 0 &\leq x_2, \\ 18 &\geq x_1 + x_2 \geq 12. \end{aligned}$$

Now the feasible set can be visualized in two dimensions,

Figure 7: Two dimensional feasible set S denoted in green.

Given the context of the problem, we are looking for some discrete answer, whether it's in a multiple of $1/8$ or 1 whole of a salmon. However we treat the problem as continuous i.e. our feasible set is continuous.

Exercise Problem P4: Complete the following classification table for the example problems:

Solution:

name	discrete	constrained	linear	quadratic	dimension
calcone				✓	1
fit			✓	✓	3
salmon	✓ (in interpretation)	✓	✓	✓	3
tsp	✓	✓			n
glacier		✓			∞